Matemáticas I
1º Bach.

SOLUCIONES DE LA FICHA · UNIDAD 2

Álgebra. Ecuaciones e inecuaciones

FentMates

Intenta resolver cada ejercicio antes de mirar su solución. Los resultados se dan de manera exacta (fracciones y raíces) y, cuando hace falta, con el valor aproximado. Puedes comprobar las divisiones, la factorización y las inecuaciones con las herramientas interactivas de los apuntes de la unidad.

1 Polinomios. Operaciones

1.

a) $\boxed{-16x^2 - 17x + 10}$

b) $\boxed{15x^2 + 3x - 3}$

c) $\boxed{6x^4 - 10x^3 - 35x^2 + 23x}$

d) $\boxed{3x^5 - 2x^4 - 20x^2 - 16x + 32}$

e) $\boxed{x^3 - \frac{9x^2}{10} + \frac{4x}{15}}$

2. $(x+y+z)^2 = \big((x+y)+z\big)^2 = (x+y)^2 + 2(x+y)z + z^2 = x^2 + 2xy + y^2 + 2xz + 2yz + z^2$. $\boxed{\text{Queda demostrada.}}$

3. $\boxed{16x^4 - 24x^3 + 25x^2 - 12x + 4}$ (con la identidad del ejercicio anterior).

4.

a) $6x^4 + (3b-10)x^3 + (2a-5b)x^2 + abx$: $3b - 10 = -19$, $ab = -6 \Rightarrow \boxed{a = 2,\ b = -3}$

b) $2x^2 + 9x + 18 - 3a$: $\boxed{b = 9}$, $18 - 3a = 3 \Rightarrow \boxed{a = 5}$

5.

a) $c(x) = \boxed{x + \frac{2}{3}}$, $r(x) = \boxed{-x - \frac{19}{3}}$

b) $c(x) = \boxed{3x^2 - 3x - 5}$, $r(x) = \boxed{10x + 14}$

c) $c(x) = \boxed{\frac{1}{2}}$, $r(x) = \boxed{-3x^2 + \frac{9}{2}}$

CC BY-NC-SA 4.0 · fentmates.org

6.

a)

	3	2	1	−5
−2		−6	8	−18
	3	−4	9	−23

$c(x) = \boxed{3x^2 - 4x + 9}$, $r = \boxed{-23}$

b)

	3	0	−2	−1	4
−2		−6	12	−20	42
	3	−6	10	−21	46

$c(x) = \boxed{3x^3 - 6x^2 + 10x - 21}$, $r = \boxed{46}$

c)

	1	−3	−1	6
$-\frac{3}{2}$		$-\frac{3}{2}$	$\frac{27}{4}$	$-\frac{69}{8}$
	1	$-\frac{9}{2}$	$\frac{23}{4}$	$-\frac{21}{8}$

$c(x) = \boxed{\frac{x^2}{2} - \frac{9x}{4} + \frac{23}{8}}$, $r = \boxed{-\frac{21}{8}}$ (cociente de Ruffini dividido entre 2)

d)

	4	0	−3	0	5
5		20	100	485	2425
	4	20	97	485	2430

$c(x) = \boxed{-4x^3 - 20x^2 - 97x - 485}$, $r = \boxed{2430}$ (cociente de Ruffini dividido entre -1)

e)

	2	0	−12	−6	0	−117
3		6	18	18	36	108
	2	6	6	12	36	−9

$c(x) = \boxed{2x^4 + 6x^3 + 6x^2 + 12x + 36}$, $r = \boxed{-9}$

7.

a) $P(-3) = a - 3 = 0 \Rightarrow \boxed{a = 3}$

b) $P(-\frac{3}{2}) = \frac{3a}{2} - \frac{3}{2} = 0 \Rightarrow \boxed{a = 1}$

c) $P(\frac{4}{3}) = \frac{16a}{9} + \frac{160}{9} = 0 \Rightarrow \boxed{a = -10}$

d) $P(5) = b + 2930 = 0 \Rightarrow \boxed{b = -2930}$

(Con el teorema del resto, $P(r)$ debe ser 0, donde r es la raíz del divisor; es lo mismo que obtener resto 0 en Ruffini.)

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

8. $M(2) = 16 - 20 + 8 - 4 = 0$: sí es divisible.

Por Ruffini:

	2	−5	4	−4
2		4	−2	4
	2	−1	2	0

$\Rightarrow N(x) = 2x^2 - x + 2$

2 Factorización de un polinomio. Teorema del resto

9.

a) $P(1) = 0$: sí **b)** $P(2) = 36$: no **c)** $P(-1) = -18$: no **d)** $P(-4) = 0$: sí **e)** $P(-3) = -44$: no

10. $P(-1) = 1 + 2 - 3 = 0$, $P(3) = 9 - 6 - 3 = 0$. No: un polinomio de grado 2 tiene como máximo dos raíces, y ya las hemos encontrado: $P(x) = (x+1)(x-3)$.

11.

a) $x = -5, x = 2, x = 3$

b) $x = -\frac{1}{2}, x = 0, x = 2$ (triple)

c) $x = -1$ (triple), $x = -\frac{2}{3}, x = \frac{1}{2}$

d) $x = -2, x = 1, x = 4$ $((x-4)(x-1)(x+2))$

e) $x = -5, x = -2, x = -1$ $((x+1)(x+2)(x+5))$

f) $x = 0, x = \frac{1}{2}, x = 1, x = 2$ (doble) $(x(x-2)^2(x-1)(2x-1))$

12.

a) $(x-3)(x-1)(x+2)$ (raíces: $x = -2, x = 1, x = 3$)

b) $(x-1)^2(x+3)$ (raíces: $x = -3, x = 1$ (doble))

c) $(x-1)(x+1)^2(2x-3)$ (raíces: $x = -1$ (doble), $x = 1, x = \frac{3}{2}$)

d) $(x-2)(2x-1)(2x+3)$ (raíces: $x = -\frac{3}{2}, x = \frac{1}{2}, x = 2$)

e) $x^2(x-2)(x+2)$ (raíces: $x = -2, x = 0$ (doble), $x = 2$)

f) $x(x+1)^2(2x-1)(3x+1)$ (raíces: $x = -1$ (doble), $x = -\frac{1}{3}, x = 0, x = \frac{1}{2}$)

g) $(x-2)(x-1)(x+1)(x+2)(x^2+1)$ (raíces: $x = -2, x = -1, x = 1, x = 2$)

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

h) $t = x^2$: $\frac{t^2}{4} + \frac{3}{4}t + \frac{9}{16} = 0 \Rightarrow \Delta = \frac{9}{16} - \frac{9}{16} = 0$, $t = -\frac{3}{2} < 0$: no tiene raíces reales. $P(x) = \boxed{\frac{1}{16}(2x^2+3)^2}$

13.

a) $P(2) = k + 8 = 0 \Rightarrow \boxed{k = -8}$

b) $P(-2) = -2k - 12 = 0 \Rightarrow \boxed{k = -6}$

c) $P(-4) = -63k + 252 = 0 \Rightarrow \boxed{k = 4}$

d) $P(3) = -6k + 36 = 0 \Rightarrow \boxed{k = 6}$

14. $P(-3) = -27 - 9(3k+2) - 3 = -48 - 27k$.

a) $-48 - 27k = 0 \Rightarrow \boxed{k = -\frac{16}{9}}$ **b)** $-48 - 27k = 15 \Rightarrow \boxed{k = -\frac{7}{3}}$

15. $P(x) = ax^2+bx+c$: $c = -3$, $a+b-3 = -6$, $9a+3b-3 = 0 \Rightarrow a = 2, b = -5$: $\boxed{P(x) = 2x^2 - 5x - 3}$

16. $P(2) = 8m - 24 - 8 + 8 = 8m - 24 = 0 \Rightarrow \boxed{m = 3}$

17. $P(2) = 0$: $4a + 2b = -2$; $P(-3) = 0$: $9a - 3b = 33$. $\boxed{a = 2, b = -5}$ (y entonces $P(x) = (x-2)(x+3)(x+1)$).

18. Teorema del resto: $P(-2) = 32 - 72 + 8 + 12 + 3m = -20 + 3m = 12 \Rightarrow \boxed{m = \frac{32}{3}}$

3 Fracciones algebraicas

19.

a) $\frac{2(x-2)(x+2)}{x(x-1)(x+2)} = \boxed{\frac{2(x-2)}{x(x-1)}}$

b) $\frac{x(x-1)(x+1)}{x(x+1)(x+2)} = \boxed{\frac{x-1}{x+2}}$

c) $\frac{x^2(x-3)(x+1)}{x^2(x-3)(x+3)} = \boxed{\frac{x+1}{x+3}}$

d) $\frac{(x-1)^3}{x(x-1)^2} = \boxed{\frac{x-1}{x}}$

e) $\frac{(x-1)^2(x+1)(2x-3)}{(x-1)(x+2)(2x-3)} = \boxed{\frac{(x-1)(x+1)}{x+2}}$

f) $\frac{(x-1)^2(x+3)(2x-1)}{(x-1)(x+3)(2x-1)} = \boxed{x-1}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

g) $\dfrac{(x-3)(x-2)(x+3)}{(x-3)(x-2)^2} = \boxed{\dfrac{x+3}{x-2}}$

h) $\dfrac{(x-2)(x-1)^3}{(x-2)^2(x-1)^2} = \boxed{\dfrac{x-1}{x-2}}$

20.

a) $\boxed{\dfrac{x+2}{(x-1)(x+1)}}$

b) $\boxed{\dfrac{3x+1}{x(x+1)}}$

c) $\boxed{\dfrac{2x-5}{4x(x-1)}}$

d) $\boxed{\dfrac{2(2x+1)}{x(x-1)(x+1)}}$

e) $\boxed{\dfrac{5x^2-3x+1}{(x-2)(x+2)}}$

f) $\boxed{\dfrac{2(x^2+x-1)}{(x-1)^2(x+1)}}$

21.

a) $\boxed{\dfrac{1}{4}}$

b) $\boxed{\dfrac{x+3}{(x-2)(x+1)}}$

c) $\boxed{\dfrac{(3x+2)(4x+1)}{(x-2)(x+2)}}$

d) $\boxed{-\dfrac{2}{3x^2}}$

e) $\boxed{-\dfrac{2x^2-3x-3}{x-1}}$

f) $\boxed{\dfrac{x^2-6x-1}{2(x+1)^2}}$

g) $\boxed{x}$

22.

a) $\boxed{\dfrac{3x^4+3x^3+3x^2+4x+2}{x(x+2)(x^2+1)}}$

b) $\boxed{\dfrac{3x(x+1)}{x^2+1}}$

c) $\boxed{-\dfrac{a+b}{b^2}}$

d) $\boxed{-\dfrac{a(b^2-b-1)}{b^2}}$

e) $\boxed{\dfrac{2(5x+6)}{(x-2)(x+2)}}$

f) $\boxed{\dfrac{(x-2)(x+5)}{x+1}}$

g) $\boxed{xy(x-y)}$

h) $\boxed{\dfrac{1}{a+x}}$

i) $\boxed{\dfrac{3x+2}{2x+1}}$

j) $\boxed{\dfrac{3x^2+3x+2}{x^2+1}}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

4 Ecuaciones polinómicas

23.

a) $\boxed{x = \frac{9}{32}}$

b) $\boxed{x = -5,\ x = 0}$

c) $\boxed{x = -\frac{\sqrt{30}}{6},\ x = \frac{\sqrt{30}}{6}}$

d) $\boxed{x = -2,\ x = \frac{11}{4}}$

e) $(x-1)(x+2)(2x+3)(3x-1) = 0$: $\boxed{x = -2,\ x = -\frac{3}{2},\ x = \frac{1}{3},\ x = 1}$

f) $(x-2)(2x+3)(x^2+3) = 0$: $\boxed{x = -\frac{3}{2},\ x = 2}$

g) $x(x+1)(2x-1)(3x+1) = 0$: $\boxed{x = -1,\ x = -\frac{1}{3},\ x = 0,\ x = \frac{1}{2}}$

h) $(x-2)(x-1)(x+3) = 0$: $\boxed{x = -3,\ x = 1,\ x = 2}$

i) $(x-3)^2(x+3) = 0$: $\boxed{x = -3,\ x = 3}$

j) $(x-3)(x+2)(x^2+1) = 0$: $\boxed{x = -2,\ x = 3}$

k) $(x-1)(x+1)(x^2+1) = 0$: $\boxed{x = -1,\ x = 1}$

l) $x(x-3)(x+1)(x+3) = 0$: $\boxed{x = -3,\ x = -1,\ x = 0,\ x = 3}$

24.

a) $t = x^2$: $t^2 - 17t + 16 = 0 \Rightarrow t = 16,\ t = 1$: $\boxed{x = \pm 4,\ x = \pm 1}$

b) $t = x^3$: $t^2 - 9t + 8 = 0 \Rightarrow t = 8,\ t = 1$: $\boxed{x = 2,\ x = 1}$

c) $t^2 - 8t - 9 = 0 \Rightarrow t = 9$ ($t = -1$ no da soluciones): $\boxed{x = \pm 3}$

d) $t = x^3$: $t^2 + 2t - 3 = 0 \Rightarrow t = 1,\ t = -3$: $\boxed{x = 1,\ x = -\sqrt[3]{3}}$

e) Desarrollando: $2x^4 + 12x^2 - 14 = 0 \Rightarrow t^2 + 6t - 7 = 0 \Rightarrow t = 1$ ($t = -7$ no da soluciones): $\boxed{x = \pm 1}$

f) Multiplicamos por x^2: $x^4 - 10x^2 + 9 = 0 \Rightarrow t = 9,\ t = 1$: $\boxed{x = \pm 3,\ x = \pm 1}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

25.

a) $\boxed{x = -2,\ x = -\sqrt{3},\ x = \sqrt{3},\ x = 2}$

b) $\boxed{x = -3,\ x = -\frac{2}{3},\ x = 0,\ x = 3,\ x = 4}$

c) $\boxed{x = -3,\ x = 2}$

d) $\boxed{x = -1,\ x = 2,\ x = \frac{7}{2}}$

5 Ecuaciones racionales

26.

a) $8x^2 + 8x + 8 = 7x^3 \Rightarrow (x-2)(7x^2 + 6x + 4) = 0$ ($\Delta < 0$): $\boxed{x = 2}$

b) Multiplicamos por $3(x-1)$: $2x^2 + 4x + 9 = 11 \Rightarrow x^2 + 2x - 1 = 0$: $\boxed{x = -1 \pm \sqrt{2}}$

c) $2x(x+2) + 3x(x-2) = 6x^2 \Rightarrow x^2 + 2x = 0 \Rightarrow x = 0,\ x = -2$; $x = -2$ anula un denominador: $\boxed{x = 0}$

d) m.c.m. $= (x-1)(x-2)(x-4)$: $3(x-1) + (x-5)(x-2) = (x-1)(x-2) \Rightarrow x^2 - 4x + 7 = x^2 - 3x + 2$: $\boxed{x = 5}$

e) Multiplicamos por $\sqrt{2}\,x$: $x^2 + 2 = 2x^2 \Rightarrow \boxed{x = \pm\sqrt{2}}$

f) $-\dfrac{x}{x-2} = -x \Rightarrow x = x(x-2) \Rightarrow x^2 - 3x = 0$: $\boxed{x = 0,\ x = 3}$

g) Simplificando: $\dfrac{2x(x+1)}{(x-1)(x+3)} = 2 \Rightarrow x^2 + x = x^2 + 2x - 3$: $\boxed{x = 3}$

h) El primer miembro es $\dfrac{x+1}{x}$: es una $\boxed{\text{identidad}}$, se cumple para todo $x \neq 0,\ x \neq -1$.

6 Ecuaciones con radicales

27.

a) $2x - 3 = (1-x)^2 \Rightarrow (x-2)^2 = 0$; $x = 2$: $-1 + 1 = 0 \neq 2$: $\boxed{\text{no tiene solución}}$

b) $2x^2 + 2x - 3 = (x+1)^2 \Rightarrow x^2 = 4$; $x = 2$ no vale ($-1 - 2 < 0$): $\boxed{x = -2}$

c) $9(7x+1) = (5x-7)^2 \Rightarrow 25x^2 - 133x + 40 = 0 \Rightarrow x = 5,\ x = \frac{8}{25}$; $x = \frac{8}{25}$ no vale: $\boxed{x = 5}$

d) $2x - 1 = 12$: $\boxed{x = \dfrac{13}{2}}$

e) $2 - 5x = 3x^2 \Rightarrow 3x^2 + 5x - 2 = 0 \Rightarrow x = -2,\ x = \frac{1}{3}$; $x = \frac{1}{3}$ no vale: $\boxed{x = -2}$

f) $2 + \sqrt{x-4} = 12 - x \Rightarrow x - 4 = (10-x)^2 \Rightarrow x^2 - 21x + 104 = 0 \Rightarrow x = 8,\ x = 13$; $x = 13$ no vale: $\boxed{x = 8}$

g) $x = 4 - 4\sqrt{x-4} + x - 4 \Rightarrow \sqrt{x-4} = 0$: $\boxed{x = 4}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

h) $x-46=-14\sqrt{x+5} \Rightarrow x^2-288x+1136=0 \Rightarrow x=4, x=284$; $x=284$ no vale: $\boxed{x=4}$

i) $10\sqrt{x-1}=20 \Rightarrow \sqrt{x-1}=2$: $\boxed{x=5}$

j) $2\sqrt{2x^2-14x}=8-2x \Rightarrow x^2-6x-16=0 \Rightarrow x=8, x=-2$; ninguna vale: no tiene solución

k) $\sqrt{1-4x}=3\sqrt{3+x} \Rightarrow 1-4x=27+9x$: $\boxed{x=-2}$

28.

a) $x^2-1=(x-1)^3 \Rightarrow x(x-1)(x-3)=0$; todas valen ($\sqrt[3]{-1}=-1$): $\boxed{x=0, x=1, x=3}$

b) $\sqrt{3x+7}=x+1 \Rightarrow x^2-x-6=0 \Rightarrow x=3, x=-2$; $x=-2$ no vale: $\boxed{x=3}$

c) $4x-1=(x-4)^3 \Rightarrow (x-7)(x^2-5x+9)=0$: $\boxed{x=7}$

d) $(4-2x)^2=8x^2-16x \Rightarrow 4x^2=16$; las dos valen: $\boxed{x=2, x=-2}$

e) $(2x+2)^2=6x+10 \Rightarrow 2x^2+x-3=0 \Rightarrow x=1, x=-\frac{3}{2}$; $x=-\frac{3}{2}$ no vale: $\boxed{x=1}$

29. $\sqrt{x+1}=2\sqrt{x-11} \Rightarrow x+1=4x-44$: $\boxed{x=15}$ (comprobación: $\sqrt{16}=4=2\sqrt{4}$).

7 Ecuaciones logarítmicas

30.

a) $2(x+3)=x+5$: $\boxed{x=-1}$

b) $15-2x=x^2 \Rightarrow x=3, x=-5$; $x=-5$ no vale: $\boxed{x=3}$

c) $\dfrac{2x+3}{x-1}=36 \Rightarrow 34x=39$: $\boxed{x=\dfrac{39}{34}}$

d) $x^2-3x+2=x^2 \Rightarrow \boxed{x=\dfrac{2}{3}}$ (vale: $x>0$ y $x^2-3x+2=\frac{4}{9}>0$)

e) $\dfrac{2x-2}{x}=\dfrac{(x-1)^2}{x} \Rightarrow (x-1)(x-3)=0$; $x=1$ no vale: $\boxed{x=3}$

f) $\log x^2=\log\dfrac{x}{20} \Rightarrow x^2=\dfrac{x}{20}$; $x=0$ no vale: $\boxed{x=\dfrac{1}{20}}$

g) $2x(4-5x)=3(2x-x^2) \Rightarrow 7x^2-2x=0$; $x=0$ no vale: $\boxed{x=\dfrac{2}{7}}$

h) $\dfrac{5(x+1)(x+2)}{x+2}=4 \Rightarrow 5x+5=4$: $\boxed{x=-\dfrac{1}{5}}$

i) $4-x=(x+2)^2 \Rightarrow x^2+5x=0$; $x=-5$ no vale: $\boxed{x=0}$

j) Cambio de base: $\dfrac{2\log 3}{\log(x+3)}=\dfrac{\log 3}{\log 5} \Rightarrow \log(x+3)=2\log 5 \Rightarrow x+3=25$: $\boxed{x=22}$

k) En base 3: $\log_3 32-\log_3(6-x)=2\log_3 x \Rightarrow x^2(6-x)=32 \Rightarrow (x+2)(x-4)^2=0$: $\boxed{x=4}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

31. $\log x + \log 2 = 1 \Rightarrow \log(2x) = \log 10$: $\boxed{x = 5}$

32. $2\log x = \log 4 + \log 9 \Rightarrow x^2 = 36$ ($x > 0$): $\boxed{x = 6}$

8 Ecuaciones exponenciales

33.

a) $2^{2x^2+2} = 2^{5x+5} \Rightarrow 2x^2 - 5x - 3 = 0$: $\boxed{x = 3,\ x = -\frac{1}{2}}$

b) $262144 = 4^9 \Rightarrow (x-2)^2 = 9$: $\boxed{x = 5,\ x = -1}$

c) $x + 4 = \frac{\log 5}{\log 3}$: $\boxed{x = \frac{\log 5}{\log 3} - 4 \approx -2{,}535}$

d) $2^{-x} = 2^{2x(x-1)} \Rightarrow 2x^2 - x = 0$: $\boxed{x = 0,\ x = \frac{1}{2}}$

e) $13^{x^2+2x} = 13^{-1} \Rightarrow (x+1)^2 = 0$: $\boxed{x = -1}$

f) $2x^2 - 4x - 2 = 4 \Rightarrow x^2 - 2x - 3 = 0$: $\boxed{x = 3,\ x = -1}$

g) $6^{4x-3} = 12 \Rightarrow 4x - 3 = \frac{\log 12}{\log 6}$: $\boxed{x = \frac{3 + \log_6 12}{4} \approx 1{,}097}$

h) $\left(\frac{3}{2}\right)^x = \frac{3}{2}$: $\boxed{x = 1}$

i) $3^{3/x} = 3^{x+2} \Rightarrow x^2 + 2x - 3 = 0$: $\boxed{x = 1,\ x = -3}$

j) $2^{x+4} = 2^{3x} \Rightarrow x + 4 = 3x$: $\boxed{x = 2}$

k) $x^2 - 5x + 6 = 0$: $\boxed{x = 2,\ x = 3}$

l) $7^{\frac{2(x-5)}{3}} = 7^{-\frac{x-1}{2}} \Rightarrow 4(x-5) = -3(x-1)$: $\boxed{x = \frac{23}{7}}$

34.

a) $2^x\left(2 + 1 + \frac{1}{2}\right) = 28 \Rightarrow 2^x = 8$: $\boxed{x = 3}$

b) $5^x\left(125 - \frac{1}{5}\right) = 3120 \Rightarrow 5^x = 25$: $\boxed{x = 2}$

c) $4^x \cdot \frac{21}{16} = 336 \Rightarrow 4^x = 256$: $\boxed{x = 4}$

d) $3^x\left(9 + 3 + 1 + \frac{1}{3}\right) = 120 \Rightarrow 3^x = 9$: $\boxed{x = 2}$

e) $2^x \cdot \frac{15}{16} = 960 \Rightarrow 2^x = 1024$: $\boxed{x = 10}$

f) $t = 6^x$: $t^2 + 8t - 9 = 0 \Rightarrow t = 1$ ($t = -9$ no vale): $\boxed{x = 0}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

35.

a) $t = 3^x$: $3t^2 - 10t + 3 = 0 \Rightarrow t = 3, t = \frac{1}{3}$: $\boxed{x = 1, x = -1}$

b) $t = 10^{x+2}$: $2t^2 + 3t - 5 = 0 \Rightarrow t = 1$ ($t = -\frac{5}{2}$ no vale) $\Rightarrow x + 2 = 0$: $\boxed{x = -2}$

c) $t = 3^x$: $t^2 + 5t - 24 = 0 \Rightarrow t = 3$ ($t = -8$ no vale): $\boxed{x = 1}$

d) $t = 3^x$: $t^2 + \frac{t^2}{3} + \frac{t}{3} = 111 \Rightarrow 4t^2 + t - 333 = 0 \Rightarrow t = 9$: $\boxed{x = 2}$

e) $t = 3^x$: $9t + \frac{t^2}{9} = 90 \Rightarrow t^2 + 81t - 810 = 0 \Rightarrow t = 9$ ($t = -90$ no vale): $\boxed{x = 2}$

f) $t = 2^x$: $\frac{t^2}{16} - \frac{5t}{8} + 1 = 0 \Rightarrow t^2 - 10t + 16 = 0 \Rightarrow t = 8, t = 2$: $\boxed{x = 3, x = 1}$

g) $t = 3^x$: $81t^2 + 27t - 810 = 0 \Rightarrow 3t^2 + t - 30 = 0 \Rightarrow t = 3$ ($t = -\frac{10}{3}$ no vale): $\boxed{x = 1}$

h) $t = 2^x$: $4t^2 + 8t - 320 = 0 \Rightarrow t^2 + 2t - 80 = 0 \Rightarrow t = 8$ ($t = -10$ no vale): $\boxed{x = 3}$

9 Problemas de ecuaciones

36. $5x + 2 \cdot 50 + 0 \cdot 50 = 4(x + 100) \Rightarrow$ $\boxed{x = 300 \text{ L}}$

37. $30 \cdot 5 + 10x = 7(30 + x) \Rightarrow 3x = 60$: $\boxed{20 \text{ L}}$

38. En 1 kg, x kg de té blanco: $18x + 14(1 - x) = 16{,}4 \Rightarrow x = 0{,}6$: $\boxed{60\,\% \text{ blanco y } 40\,\% \text{ verde}}$

39. El 10 % son 15: $\boxed{150 \text{ alumnos}}$

40. A la venta queda el $75\,\%$ del $20\,\%$, es decir, el $15\,\%$: $0{,}15x = 18\,000 \Rightarrow$ $\boxed{120\,000 \text{ entradas}}$

41. África es el 15 %: $0{,}15x = 3 \Rightarrow$ $\boxed{20 \text{ millones de euros}}$

42. $5x - 2(20 - x) = 58 \Rightarrow 7x = 98$: $\boxed{14 \text{ aciertos y } 6 \text{ errores}}$

43. $(x - 20)\left(\frac{125}{x} + 0{,}4\right) = 147 \Rightarrow x^2 - 75x - 6250 = 0 \Rightarrow x = 125$ ($x = -50$ no tiene sentido): $\boxed{125 \text{ kg}}$

44. $x^2 + (x + 1)^2 = 25 \Rightarrow x^2 + x - 12 = 0 \Rightarrow x = 3$: $\boxed{3 \text{ cm y } 4 \text{ cm}}$

45. $x^2 + (x + 2)^2 = (x + 4)^2 \Rightarrow x^2 - 4x - 12 = 0 \Rightarrow x = 6$: $\boxed{6, 8 \text{ y } 10 \text{ cm}}$

46. $x + y = 70, x^2 + y^2 = 2500 \Rightarrow x^2 - 70x + 1200 = 0$: $\boxed{40 \text{ m} \times 30 \text{ m}}$

47. $\frac{1}{5} + \frac{1}{20} = \frac{1}{4}$: $\boxed{T = 4 \text{ h}}$

48. $\frac{1}{2} + \frac{1}{3} = \frac{5}{6} \Rightarrow T = \frac{6}{5}$ h: $\boxed{1 \text{ h } 12 \text{ min}}$

49. $\frac{1}{t} + \frac{1}{2t} = \frac{1}{10} \Rightarrow t = 15$: $\boxed{15 \text{ h y } 30 \text{ h}}$

50. $\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40} + \frac{1}{50} = \frac{137}{600} \Rightarrow T = \frac{600}{137} \approx$ $\boxed{4{,}38 \text{ h} \approx 4 \text{ h } 23 \text{ min}}$

51. $\frac{600}{x - 10} = \frac{600}{x} + 5 \Rightarrow x^2 - 10x - 1200 = 0 \Rightarrow x = 40$: $\boxed{40 \text{ personas}}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

52.

a) $\boxed{y(0) = 300 \text{ enfermos}}$

b) $200e^{0{,}2t} = 800 \Rightarrow t = 5\ln 4 \approx 6{,}93$: $\boxed{\text{el 7 de enero (hacia el final del día)}}$

53.

a) $e^{-0{,}1t} = \frac{1}{2} \Rightarrow t = 10\ln 2 \approx \boxed{6{,}93 \text{ días}}$

b) $27e^{-1} \approx \boxed{9{,}93 \text{ mg}}$

54. $2^t = \dfrac{100}{9} \Rightarrow t = \dfrac{\log(100/9)}{\log 2} \approx \boxed{3{,}47 \text{ h} \approx 3 \text{ h } 28 \text{ min}}$

55.

a) $1 + Ka = 4,\ 1 + Ka^2 = 2{,}5 \Rightarrow Ka = 3,\ Ka^2 = 1{,}5$: $\boxed{a = \dfrac{1}{2},\ K = 6}$

b) $1 + 6 \cdot 0{,}5^{x/10} = \frac{4}{3} \Rightarrow 0{,}5^{x/10} = \frac{1}{18} \Rightarrow x = 10\log_2 18 \approx \boxed{41{,}7 \text{ años}}$

10 Inecuaciones con una incógnita

56.

a) $-8 + 4 - 2 = -6 \leq 6$: $\boxed{\text{sí}}$

b) $\frac{1}{e} - 1 \approx -0{,}63 \not\geq 0$: $\boxed{\text{no}}$

c) $2^{-0{,}5} - 2{,}5 \approx -1{,}79 < 3^{-0{,}5} \approx 0{,}58$: $\boxed{\text{sí}}$

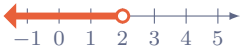
57.

a) $4x - 12 - x + 2 \leq 4x \Rightarrow -x \leq 10$: $\boxed{x \in [-10,\ +\infty)}$



b) $9x - 18 > 9x + 1 \Rightarrow -18 > 1$, falso: $\boxed{\text{no tiene solución}}$

c) $6x + 8 < \frac{x+38}{2} \Rightarrow 11x < 22$: $\boxed{x \in (-\infty, 2)}$



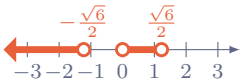
d) $2x^2 - 8x \leq 0 \Rightarrow 2x(x-4) \leq 0$: $\boxed{x \in [0, 4]}$



Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

58.

a) $-x(2x^2-3)>0$: $\boxed{x\in(-\infty,-\frac{\sqrt{6}}{2})\cup(0,\frac{\sqrt{6}}{2})}$



b) $(x+3)(3x-8)\geq 0$: $\boxed{x\in(-\infty,-3]\cup[\frac{8}{3},+\infty)}$



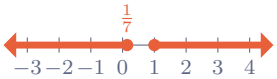
c) $x(x-2)(x+2)>0$: $\boxed{x\in(-2,0)\cup(2,+\infty)}$



d) $(x-1)(x+1)^2\leq 0$: $\boxed{x\in(-\infty,1]}$



e) $(x-1)(7x-1)\geq 0$: $\boxed{x\in(-\infty,\frac{1}{7}]\cup[1,+\infty)}$



f) $-(x-1)(x+1)(x+2)\leq 0$: $\boxed{x\in[-2,-1]\cup[1,+\infty)}$



g) $(x-1)(x+1)(x^2+1)>0$: $\boxed{x\in(-\infty,-1)\cup(1,+\infty)}$

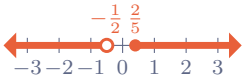


h) $(x+1)(x+2)<0$: $\boxed{x\in(-2,-1)}$



59.

a) $\dfrac{5x-2}{2x+1}\geq 0$: $\boxed{x\in(-\infty,-\frac{1}{2})\cup[\frac{2}{5},+\infty)}$



b) $\dfrac{(x-2)(x+1)}{(x-1)(x+2)}\geq 0$: $\boxed{x\in(-\infty,-2)\cup[-1,1)\cup[2,+\infty)}$



Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

c) $\dfrac{4x-5}{(x+1)(4x-5)} < 0$: $\boxed{x \in (-\infty, -1)}$



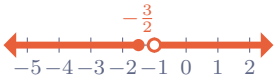
d) $\dfrac{x^3}{x-9} \geq 0$: $\boxed{x \in (-\infty, 0] \cup (9, +\infty)}$



e) $\dfrac{(x-1)(x+5)}{x(x-1)(x+1)} \geq 0$: $\boxed{x \in [-5, -1) \cup (0, 1) \cup (1, +\infty)}$



f) $\dfrac{2x+3}{(x+1)(x^2-x+1)} \geq 0$: $\boxed{x \in (-\infty, -\frac{3}{2}] \cup (-1, +\infty)}$



11 Problemas de inecuaciones

60. $10 + 0{,}05x < 0{,}12x \Rightarrow x > \frac{1000}{7} \approx 142{,}9$: $\boxed{\text{a partir de 143 minutos}}$

61. $1{,}2 + 0{,}7x \geq 5 \Rightarrow x \geq \frac{38}{7} \approx 5{,}43$: $\boxed{x \in [\frac{38}{7}, 10]}$



62. $20t - 5t^2 > 15 \Rightarrow t^2 - 4t + 3 < 0 \Rightarrow t \in (1{,}3)$: $\boxed{2 \text{ segundos}}$

12 Ampliación y problemas

63. $P(-1) = 1 + 2 + 1 = \boxed{4}$

64. $P(x) = a(x-1)(x+2)(x-3)$; $P(0) = 6a = 12 \Rightarrow a = 2$: $\boxed{2x^3 - 4x^2 - 10x + 12}$

65.

a) $\dfrac{(x-3)(x^2+3x+9)}{(x-3)(x+3)} = \boxed{\dfrac{x^2+3x+9}{x+3}}$

b) $\dfrac{\frac{x-1}{x}}{\frac{(x-1)(x+1)}{x^2}} = \boxed{\dfrac{x}{x+1}}$

66. $2x^2 + 2(x+1)^2 = 5x(x+1) \Rightarrow x^2 + x - 2 = 0$: $\boxed{x = 1, x = -2}$ (ninguna anula los denominadores).

67. $x + 3 = 1 + 2\sqrt{x-2} + x - 2 \Rightarrow \sqrt{x-2} = 2$: $\boxed{x = 6}$ $(3 - 2 = 1\ \checkmark)$.

68. $t = 2^x$: $2t^2 - 5t + 2 = 0 \Rightarrow t = 2, t = \frac{1}{2}$: $\boxed{x = 1, x = -1}$

69. $x^2 - 1 = 8 \Rightarrow x = \pm 3$; $x = -3$ no vale: $\boxed{x = 3}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)

70. $x^2 + 1 > 0$ siempre, así que solo hace falta $x^2 - 4 \leq 0$: $\boxed{x \in [-2, 2]}$



71. $5000 \cdot 1{,}04^t \geq 8000 \Rightarrow t \geq \dfrac{\log 1{,}6}{\log 1{,}04} \approx 11{,}98$: $\boxed{12 \text{ años}}$

Licencia Creative Commons Atribución-NoComercial-CompartirIgual 4.0 Internacional (CC BY-NC-SA 4.0)